1	The energy E stored in a certain electronic component is given by $$E = \frac{Q^n}{2k}$$ where Q is the total charge in the component, n is an unknown integer and k is a physical quantity with SI base units $A^2\ s^4\ kg^{-1}\ m^{-2}$. What is the value of n?						
	A	-2	**B**	-1	**C**	1	**D**	2
	Answer: D $[E] = \frac{[Q]^n}{[k]}$ $kg\ m^2\ s^{-2} = \frac{A^n\ s^n}{A^2\ s^4\ kg^{-1}\ m^{-2}}$ $n = 2$						

2 A tennis ball is released from rest at the top of a tall building.